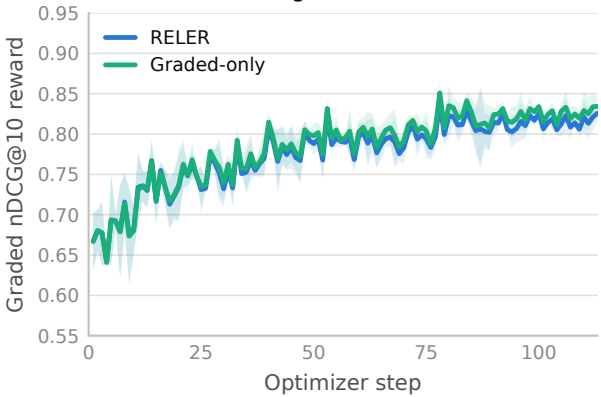
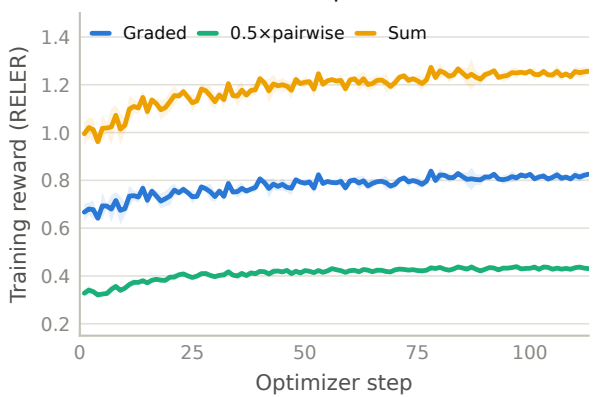

(a) Graded training reward
RELER
Graded-only
Graded nDCG@10 reward
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0
25
50
75
100
Optimizer step


(b) RELER reward components
Graded
0.5×pairwise
Sum
Training reward (RELER)
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0
25
50
75
100
Optimizer step